

# Technology Company Performance Analysis

## Project Overview:

- Technology companies maintain open-source repositories on GitHub to collaborate with developers, share software and promote innovation.
- This project analyzes the performance of GitHub repositories from leading technology companies using Python and the GitHub REST API.
- The repository data was collected, cleaned and transformed through data cleaning, feature engineering and preprocessing techniques.
- The analysis evaluated key repository metrics, including Stars, Forks, Watchers, Open Issues, Programming Language, Repository Age and Repository Status.
- Exploratory Data Analysis (EDA), statistical analysis and data visualization were performed using Pandas, NumPy, Matplotlib and Seaborn.
- Various visualizations, including bar charts, histograms, box plots, scatter plots, pie charts and correlation heatmaps, were created to identify trends, relationships and developer engagement patterns.
- The findings provide valuable insights into repository popularity, technology adoption and repository maintenance, supporting data-driven decision-making across leading technology companies.

## Data Source:

| Attribute              | Details                                                                                             |
|------------------------|-----------------------------------------------------------------------------------------------------|
| Dataset Name           | Technology Company Performance Analysis Dataset                                                     |
| Source                 | GitHub REST API (Public Open-Source Repository Data)                                                |
| Data Collection Method | Collected from the GitHub REST API using Personal Access Token (PAT) authentication and pagination. |
| Location               | Public GitHub repositories of Microsoft, Google, Oracle, IBM, SAP, Salesforce, Snowflake and SAS    |
| File Type              | CSV                                                                                                 |
| Number of Records      | 1,586 Repositories                                                                                  |
| Number of Attributes   | 13                                                                                                  |
| Domain                 | Information Technology / Open-Source Software Analytics                                             |
| Timeline               | Repository data collected from 2021–2026                                                            |

**Problem Statement:**

- 🌐 Leading technology companies maintain thousands of open-source repositories on GitHub, generating large volumes of repository data.
- 🌐 Comparing repository performance, developer engagement and technology adoption across companies is difficult without systematic analysis.
- 🌐 Organizations need data-driven insights to identify popular repositories, programming language trends and repository maintenance activities.
- 🌐 Without proper analysis, valuable information about open-source performance and developer contributions remains underutilized.
- 🌐 This project addresses these challenges by analyzing GitHub repository data to generate meaningful insights that support data-driven decision-making.

**Objectives:**

- 🌐 To evaluate the open-source repository performance of leading technology companies.
- 🌐 To compare repository popularity and developer community engagement across companies.
- 🌐 To identify programming language trends and technology adoption patterns.
- 🌐 To assess repository maintenance and project activity using key performance indicators.
- 🌐 To examine repository growth trends through creation and update timelines.
- 🌐 To provide data-driven business insights using exploratory data analysis and interactive dashboards.

**Attribute Details:**

| Attribute Name       | Description                                                          |
|----------------------|----------------------------------------------------------------------|
| Company              | Name of the technology company that owns the GitHub repository.      |
| Repository Name      | Name of the GitHub repository.                                       |
| Programming Language | Primary programming language used in the repository.                 |
| Stars                | Number of GitHub stars received, representing repository popularity. |
| Forks                | Number of times the repository has been forked by developers.        |
| Watchers             | Number of users watching the repository for updates.                 |
| Open Issues          | Number of unresolved issues currently available in the repository.   |
| Repository Size (KB) | Storage size of the repository in kilobytes.                         |
| License              | Software license associated with the repository.                     |
| Archived             | Indicates whether the repository is archived or actively maintained. |

|                |                                                         |
|----------------|---------------------------------------------------------|
| Created Date   | Date on which the repository was originally created.    |
| Updated Date   | Most recent date when repository activity was recorded. |
| Repository URL | Direct GitHub URL of the repository.                    |

### Tools & Technologies:

| Category                 | Technologies / Tools                                                                                       |
|--------------------------|------------------------------------------------------------------------------------------------------------|
| Programming Language     | Python                                                                                                     |
| Development Environment  | Google Colab                                                                                               |
| Data Collection          | GitHub REST API                                                                                            |
| API Library              | Requests                                                                                                   |
| Data Processing          | Pandas, NumPy                                                                                              |
| Data Visualization       | Matplotlib, Seaborn                                                                                        |
| Dataset Format           | CSV                                                                                                        |
| Visualization Techniques | Bar Chart, Histogram, Box Plot, Pie Chart, Scatter Plot, Correlation Heatmap, Pair Plot, Grouped Bar Chart |

### Data Pre-Processing and Feature Engineering :

#### Data Cleaning Steps

- 🌐 Verified the dataset structure using `df.info()`.
- 🌐 Examined the first and last records of the dataset.
- 🌐 Checked the number of rows and columns.
- 🌐 Generated descriptive statistics for numerical variables.
- 🌐 Identified missing values in every column.
- 🌐 Checked duplicate records.
- 🌐 Verified data types for all attributes.
- 🌐 Filled missing values in the Programming Language column with "Unknown".
- 🌐 Filled missing values in the License column with "No License".
- 🌐 Converted Created Date and Updated Date into Date Time format.
- 🌐 Renamed column names into a standardized naming convention.
- 🌐 Standardized company names.
- 🌐 Verified the dataset after cleaning.

No duplicate records were found in the dataset, indicating good data quality. Missing values were successfully handled without removing any repositories.

### Feature Engineering :

| Feature                     | Description                                                                                                                                                                |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Repository_Age_Years        | Created by calculating the difference between the repository creation date (Created_Date) and the last updated date (Updated_Date) to measure the repository age in years. |
| Repository_Status           | Created by transforming the original Archived column into a categorical feature with two values: Active and Archived to simplify repository status analysis.               |
| Repository_Popularity_Score | Created by combining Stars, Forks and Watchers using the formula Stars + Forks + Watchers to represent the overall popularity and developer engagement of each repository. |

### Exploratory Data Analysis (EDA) :

- 🌐 Displayed the first five records.
- 🌐 Displayed the last five records.
- 🌐 Examined dataset dimensions.
- 🌐 Verified column names.
- 🌐 Checked Missing Values
- 🌐 Reviewed data types.

### Missing Value Analysis

Missing values were identified for:

- 🌐 Programming Language
- 🌐 License

These missing values were replaced with meaningful labels to avoid information loss during analysis.

### Duplicate Record Validation

Duplicate records were examined using the duplicate detection function.

## Result:

No duplicate repositories were identified.

## Descriptive Statistics

Descriptive statistics were generated for numerical variables, including:

- 🌐 Stars
- 🌐 Forks
- 🌐 Watchers
- 🌐 Open Issues
- 🌐 Repository Size

These statistics provide information about average values, spread, minimum values, maximum values, and overall distribution.

## Statistical Analysis:

### Measures of Central Tendency

Measures of central tendency were calculated company-wise using the 'Stars' column. The Mean, Median and Mode help describe the average repository popularity and identify the overall distribution of repository stars.

Statistical Measures Used:

- 🌐 Mean
- 🌐 Median
- 🌐 Mode

## Interpretation

- 🌐 The statistical analysis was performed on 1,586 repositories collected from 8 leading technology companies.
- 🌐 Google recorded the highest average repository stars (Mean = 1816.86), followed by Microsoft (Mean = 1025.90), indicating stronger developer engagement.
- 🌐 SAS recorded the lowest average repository stars (Mean = 23.96), showing comparatively lower community popularity.
- 🌐 For every company, the Mean is considerably higher than the Median, indicating that a small number of highly popular repositories significantly increase the average.



- 🌐 Microsoft recorded the highest maximum repository stars (109,490), while Google recorded the second highest (51,512).
- 🌐 Overall, the repository star distribution is positively (right) skewed, meaning most repositories receive relatively few stars while only a small number become extremely popular.

### **Measures of Dispersion**

To understand the spread of repository popularity, Variance were calculated.

Statistical Measures Used:

Variance

### **Interpretation**

- 🌐 Microsoft recorded the highest Standard Deviation (7359.61), indicating extremely high variation in repository popularity.
- 🌐 Google recorded the second highest Standard Deviation (5176.96), showing that repository popularity varies considerably across projects.
- 🌐 Oracle (1558.75) and IBM (1122.53) also showed large variability.
- 🌐 SAP (216.97), Salesforce (516.40), Snowflake (155.06) and SAS (51.82) displayed comparatively lower variation.
- 🌐 Higher Standard Deviation values indicate that a few repositories receive exceptionally high community attention while most repositories receive relatively fewer stars.
- 🌐 These measures confirm that repository popularity is highly uneven across all selected technology companies.

## Distribution Analysis :

### Chart Type

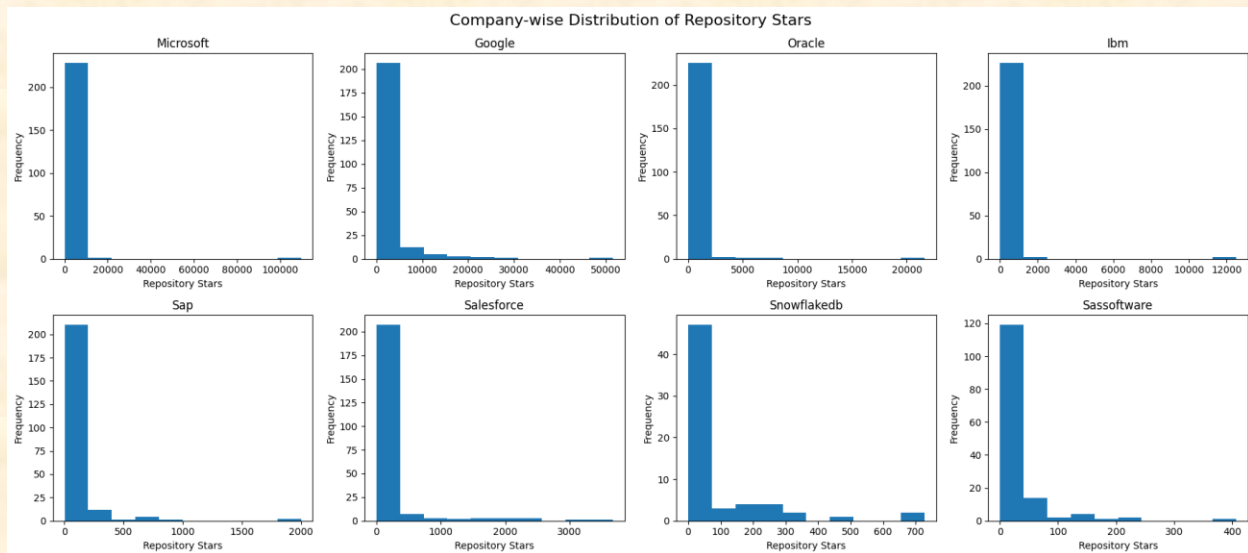
Histogram

### Columns Used

Stars

### Relationship

Repository Stars ↔ Frequency



### Interpretation

- 🌐 Microsoft has 230 repositories, with repository stars ranging from 4 to 109,490.
- 🌐 Google has 230 repositories, with stars ranging from 2 to 51,512.
- 🌐 Oracle repositories range from 0 to 21,622 while IBM repositories range from 0 to 12,500.
- 🌐 In every company, most repositories are concentrated within the lowest star ranges.
- 🌐 The histogram clearly shows a right-skewed distribution, indicating that only a few repositories receive exceptionally high stars.
- 🌐 Overall, repository popularity is highly concentrated among a small number of repositories, while the majority receive relatively limited community attention.

## Outlier Analysis:

### Chart Type

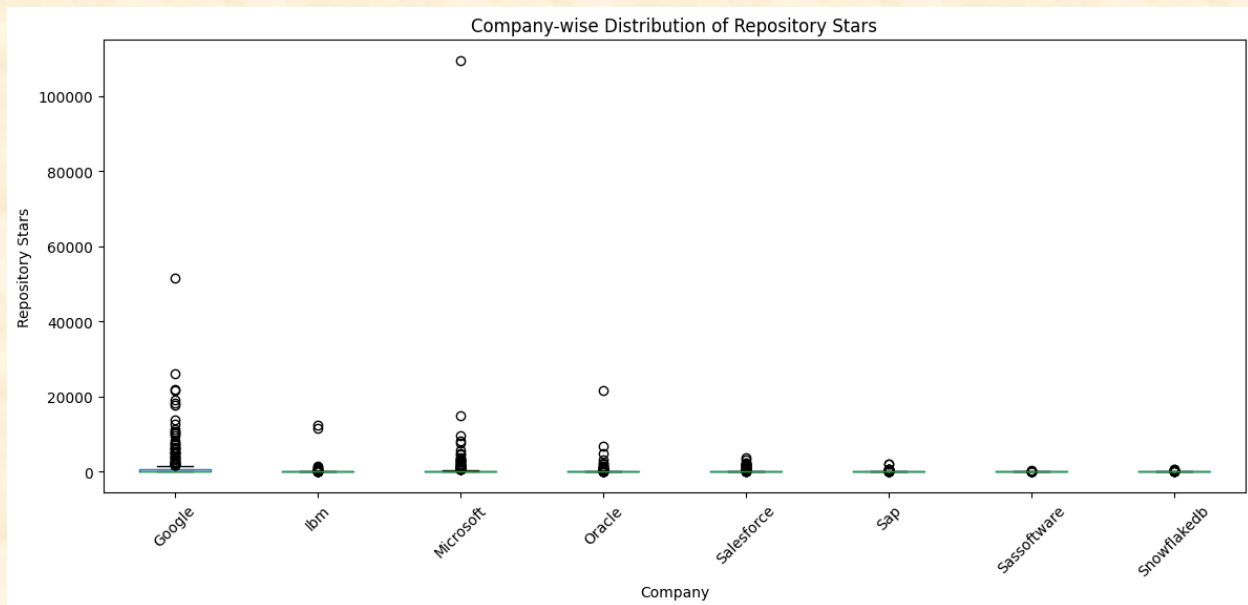
Box Plot

### Columns Used

Stars

### Relationship:

Company ↔ Repository Stars



### Interpretation

- 🌐 Google contains 43 outliers, while Microsoft contains 42 outliers, representing repositories with exceptionally high popularity.
- 🌐 Oracle contains 39 outliers, SAP 37, Salesforce 31, IBM 24, SAS 24, and Snowflake 12.
- 🌐 Google recorded the highest Median (113) among all companies.
- 🌐 Microsoft recorded the highest Maximum Stars (109,490).
- 🌐 These outliers were retained because they represent real GitHub repositories rather than data errors.
- 🌐 The box plot confirms that repository popularity is highly uneven across technology companies.



## Data Visualization

### Univariate Analysis

#### 1. Top 15 Programming Languages Used in Repositories

##### Chart Type

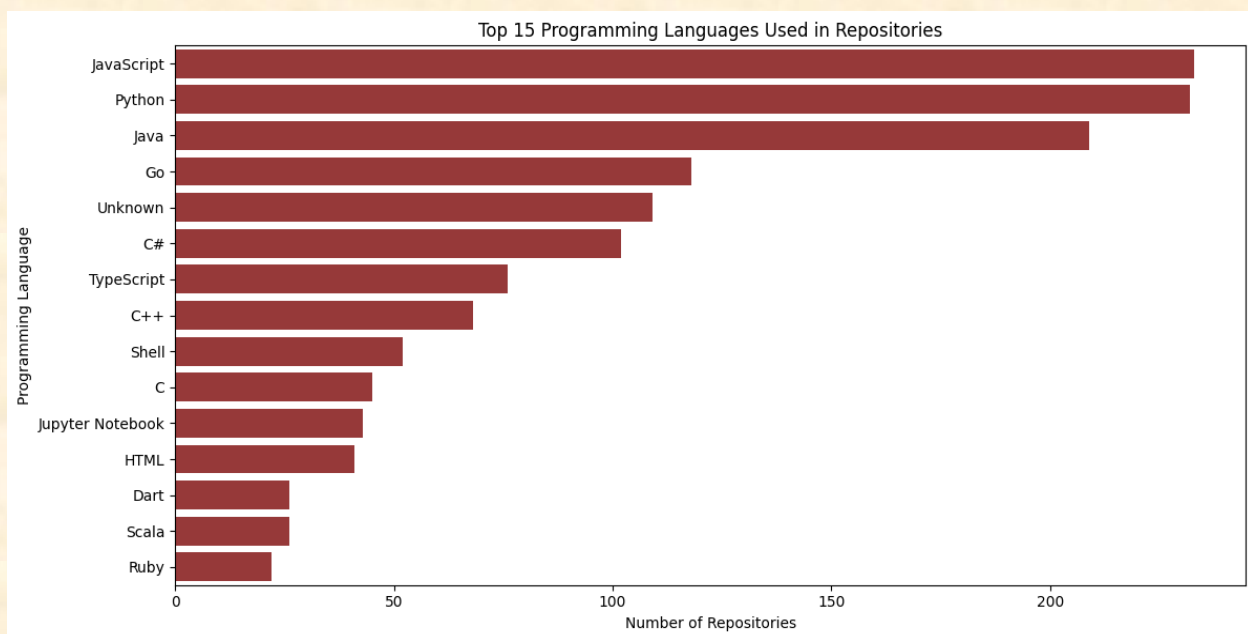
Horizontal Bar Plot

##### Columns Used

Programming\_Language

##### Relationship

Programming Language ↔ Number of Repositories



##### Interpretation

- 🌐 JavaScript is the most frequently used language with approximately 230 repositories, followed by Python (approximately 225 repositories) and Java (approximately 210 repositories).
- 🌐 Go appears in approximately 130 repositories, while the remaining languages have comparatively lower counts.
- 🌐 JavaScript, Python, and Java dominate the selected repositories, indicating their widespread adoption among leading technology companies. The visualization highlights technology adoption trends and developer preferences in open-source software development.

## 2. Distribution of Repository Age

### Chart Type

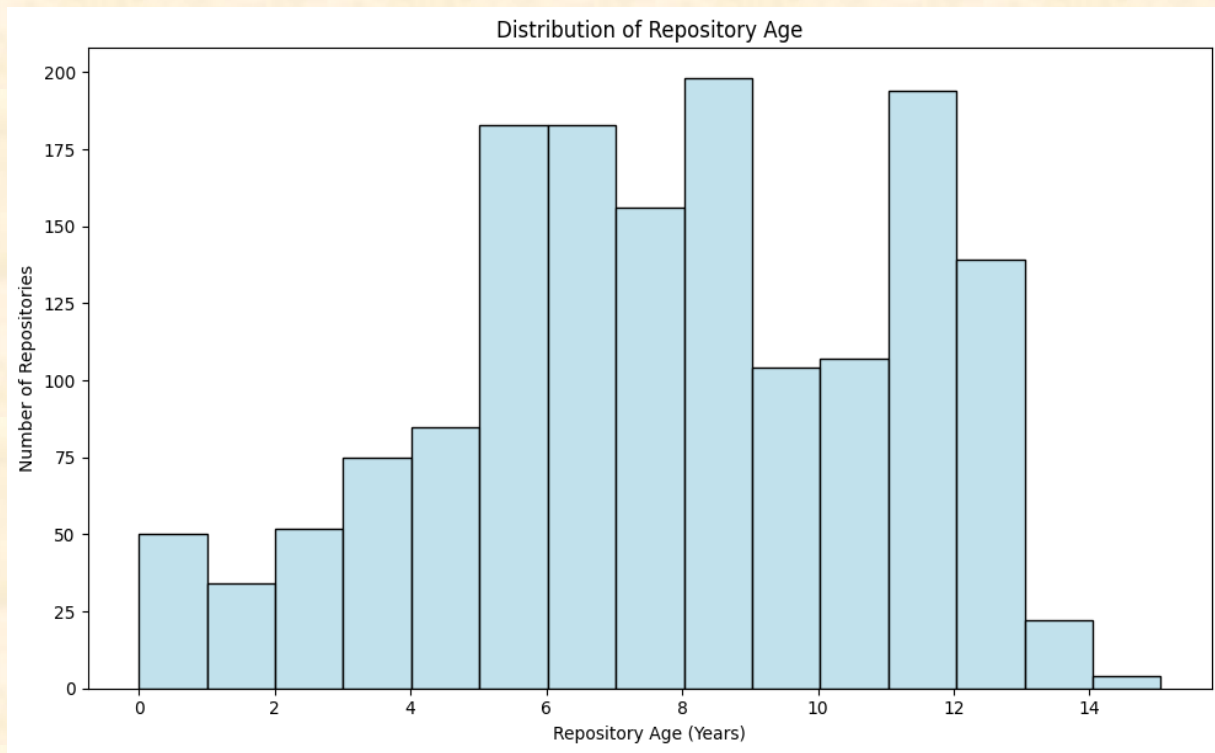
Histogram

### Columns Used

Repository\_Age\_Years

### Relationship

Repository Age  $\leftrightarrow$  Number of Repositories



### Interpretation

- 🌐 The histogram represents the age distribution of 1,586 repositories.
- 🌐 Repository age ranges from 0.00 years to 15.05 years.
- 🌐 Most repositories are approximately 5–12 years old.
- 🌐 The highest concentration occurs around 8–9 years and 11–12 years.
- 🌐 The dataset mainly contains mature and well-established repositories rather than newly created projects.
- 🌐 Repository age helps evaluate repository maturity and supports further analysis of repository popularity.

### 3.Distribution of Open Issues

#### Chart type

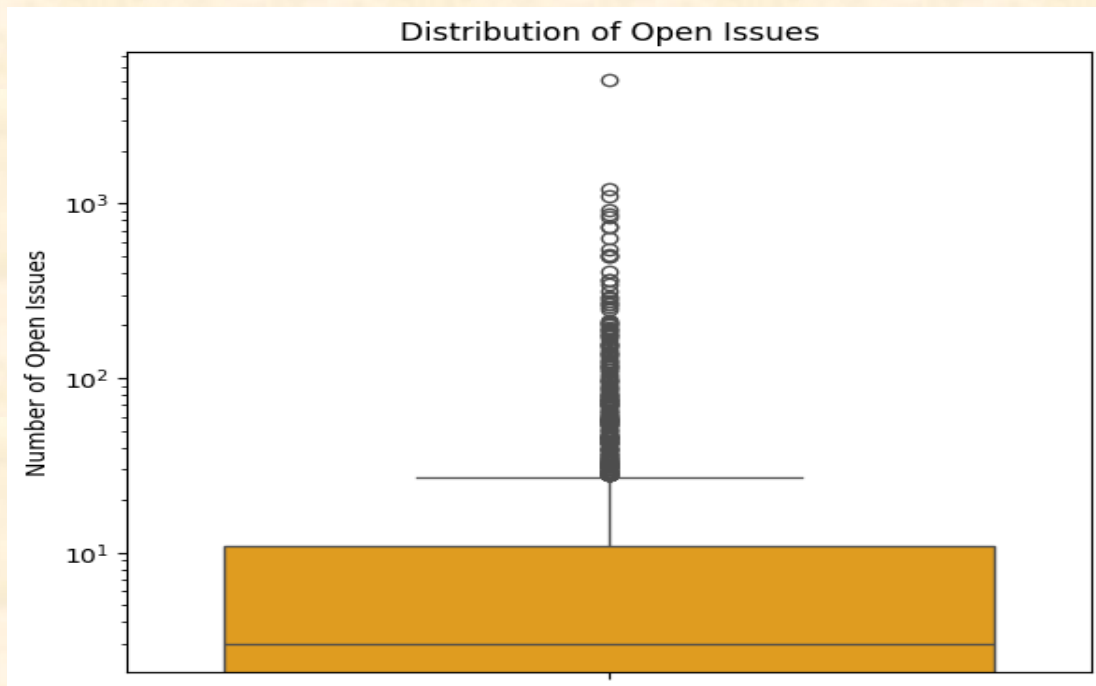
Boxplot

#### Column used

Open\_Issues

#### Relationship

Open\_Issues ↔ Repository Maintenance and Project Activity



#### Interpretation

- 🌐 The box plot shows the distribution of Open Issues across the analyzed repositories.
- 🌐 The median number of Open Issues is approximately 3, with the middle 50% of repositories having between 1 and 11 open issues.
- 🌐 The upper whisker extends to approximately 27 open issues, and repositories with values above this are identified as potential outliers.
- 🌐 Several repositories have exceptionally high numbers of open issues, exceeding 100, 1,000 and even approximately 5,000, indicating the presence of extreme outliers.
- 🌐 Overall, the distribution is highly right-skewed, showing that most repositories have relatively few open issues, while only a small number have exceptionally high numbers of open issues.

## 4.Repository Status Distribution

### Chart Type

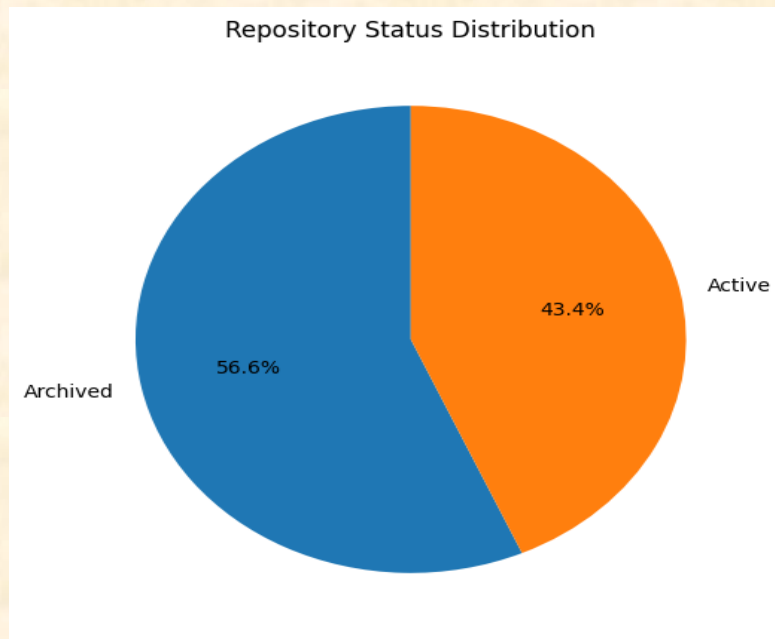
Pie Chart

### Columns Used

Repository\_Status

### Relationship

Repository Status ↔ Percentage Distribution



### Interpretation

- 🌐 The pie chart classifies repositories into Archived and Active categories.
- 🌐 Archived repositories account for 56.6% of the dataset.
- 🌐 Active repositories account for 43.4%.
- 🌐 Archived repositories exceed active repositories by 13.2 percentage points.
- 🌐 The higher percentage of archived repositories indicates that many repositories have completed development or are no longer actively maintained.
- 🌐 Repository status provides valuable insights into repository maintenance and project lifecycle across technology companies.

## Bivariate Analysis

### 1.Average Repository Stars by Company

#### Chart Type

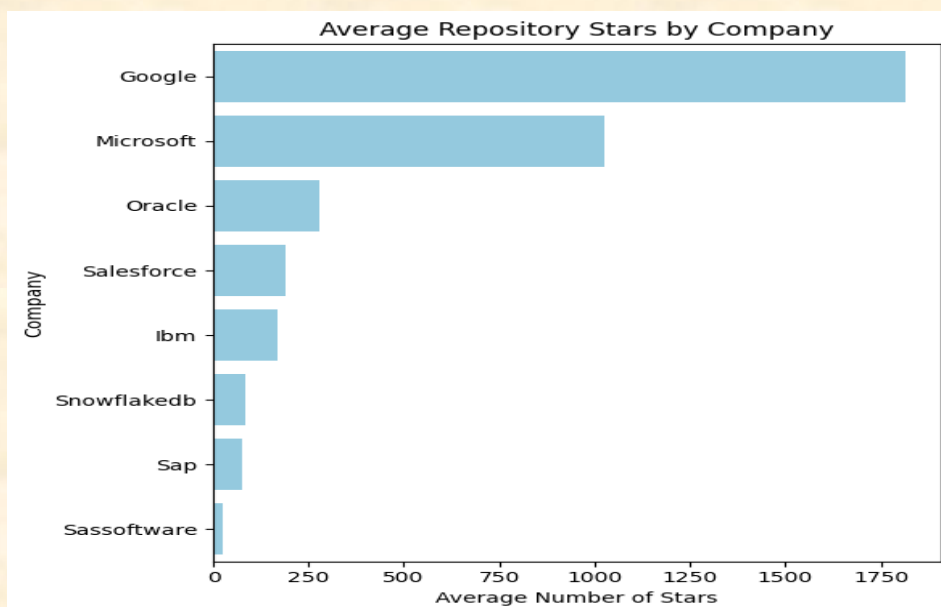
Horizontal Bar Plot

#### Columns Used

- Company
- Stars

#### Relationship

Company ↔ Average Repository Stars



#### Interpretation

- 🌐 Google recorded the highest average repository stars (1,816.86), followed by Microsoft (1,025.90).
- 🌐 Salesforce (188.68), IBM (167.88), Snowflake (83.37), SAP (77.68), Oracle (276.73) and SAS (23.96) recorded comparatively lower average repository stars.
- 🌐 The chart clearly shows that Google and Microsoft repositories receive significantly higher community attention than repositories from the remaining companies.
- 🌐 This analysis helps compare repository popularity, developer engagement and open-source performance across leading technology companies.



## 2. Relationship Between Repository Age and Stars

### Chart Type

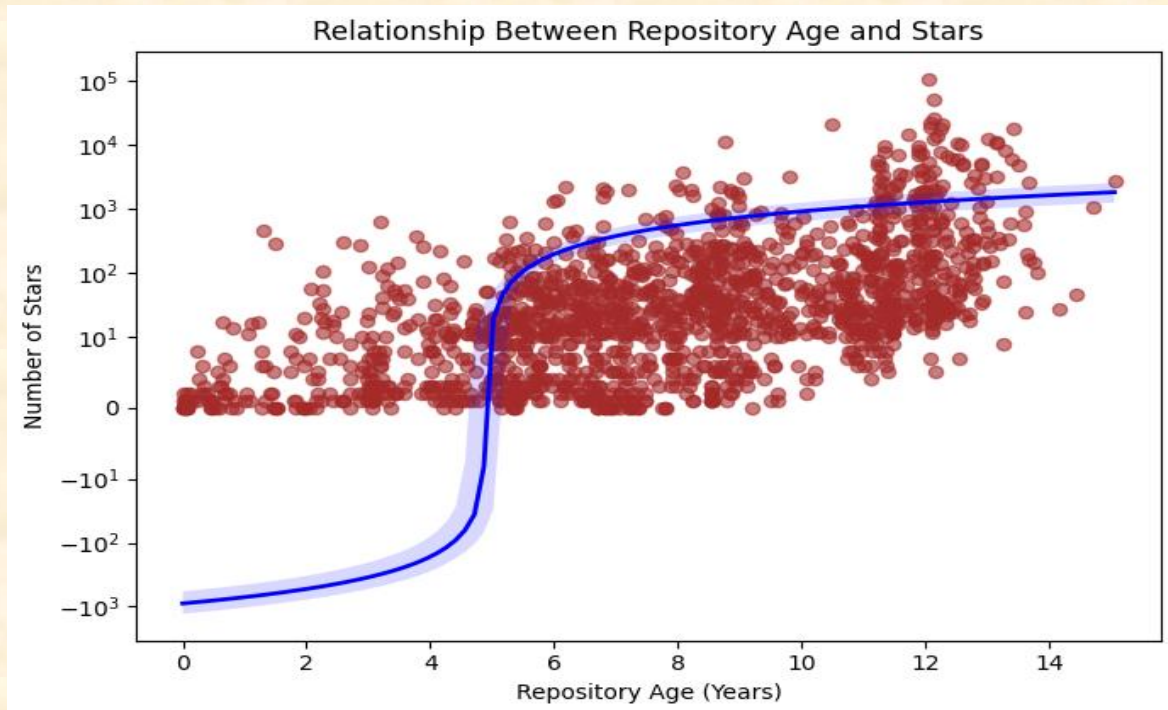
Scatter Plot with Regression Line

### Columns Used

- Repository\_Age\_Years
- Stars

### Relationship

Repository Age  $\leftrightarrow$  Repository Stars



### Interpretation

- 🌐 The scatter plot analyzes 1,586 repositories, with repository ages ranging from 0.00 to 15.05 years and repository stars ranging from 0 to 109,490.
- 🌐 The correlation coefficient ( $r = 0.1671$ ) indicates a weak positive relationship between repository age and repository stars.
- 🌐 The coefficient of determination ( $R^2 = 0.0279$ ) shows that repository age explains only 2.79% of the variation in repository popularity.
- 🌐 The regression slope (181.3859) suggests that older repositories may receive slightly more stars, and the p-value (0.000000) indicates the relationship is statistically significant.
- 🌐 Overall, repository age has limited influence on popularity, while factors such as project quality, developer adoption and community engagement play a greater role.

### 3. Average Repository Stars by Programming Language

#### Chart Type

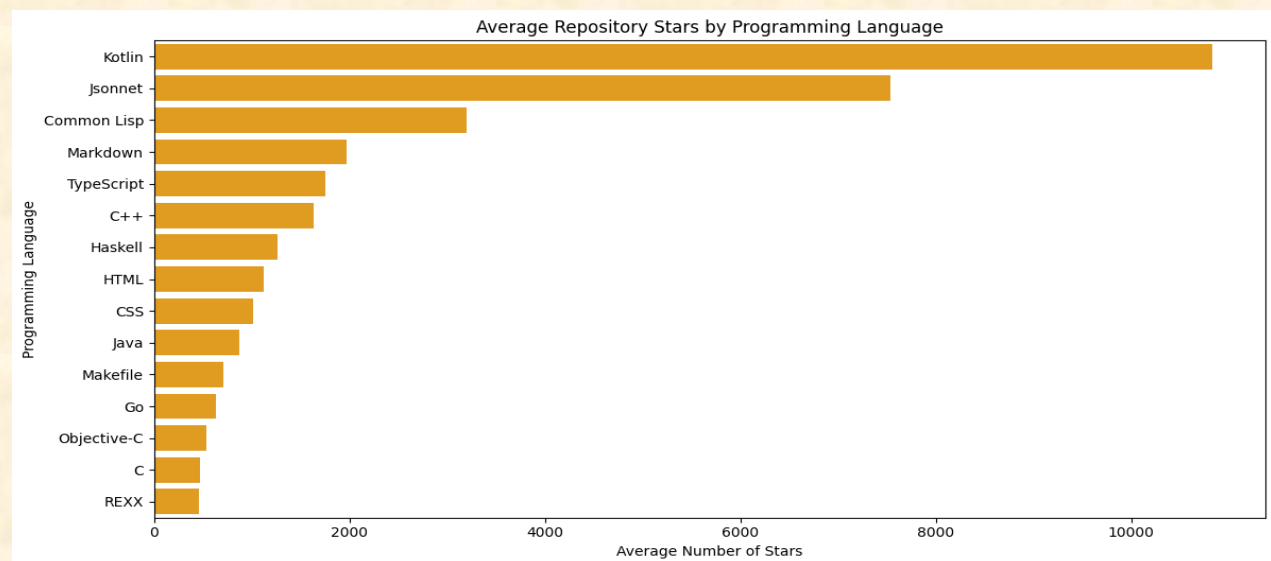
Horizontal Bar Plot

#### Columns Used

- Programming\_Language
- Stars

#### Relationship

Programming Language ↔ Average Repository Stars



#### Interpretation

- 🌐 The chart compares the average repository stars across the Top 15 programming languages.
- 🌐 Kotlin recorded the highest average repository stars (approximately 11,000), followed by Jsonnet (approximately 5,500).
- 🌐 Programming languages such as JavaScript, Python, Java, Go, TypeScript, and C# also received strong community attention, while C, REXX, and Objective-C recorded comparatively lower average stars.
- 🌐 The X-axis represents the Average Number of Stars, while the Y-axis represents the Programming Languages.
- 🌐 The visualization highlights technology adoption trends by identifying programming languages associated with highly popular repositories and higher community engagement.

Multivariate Analysis

1.Correlation Heatmap

Chart Type

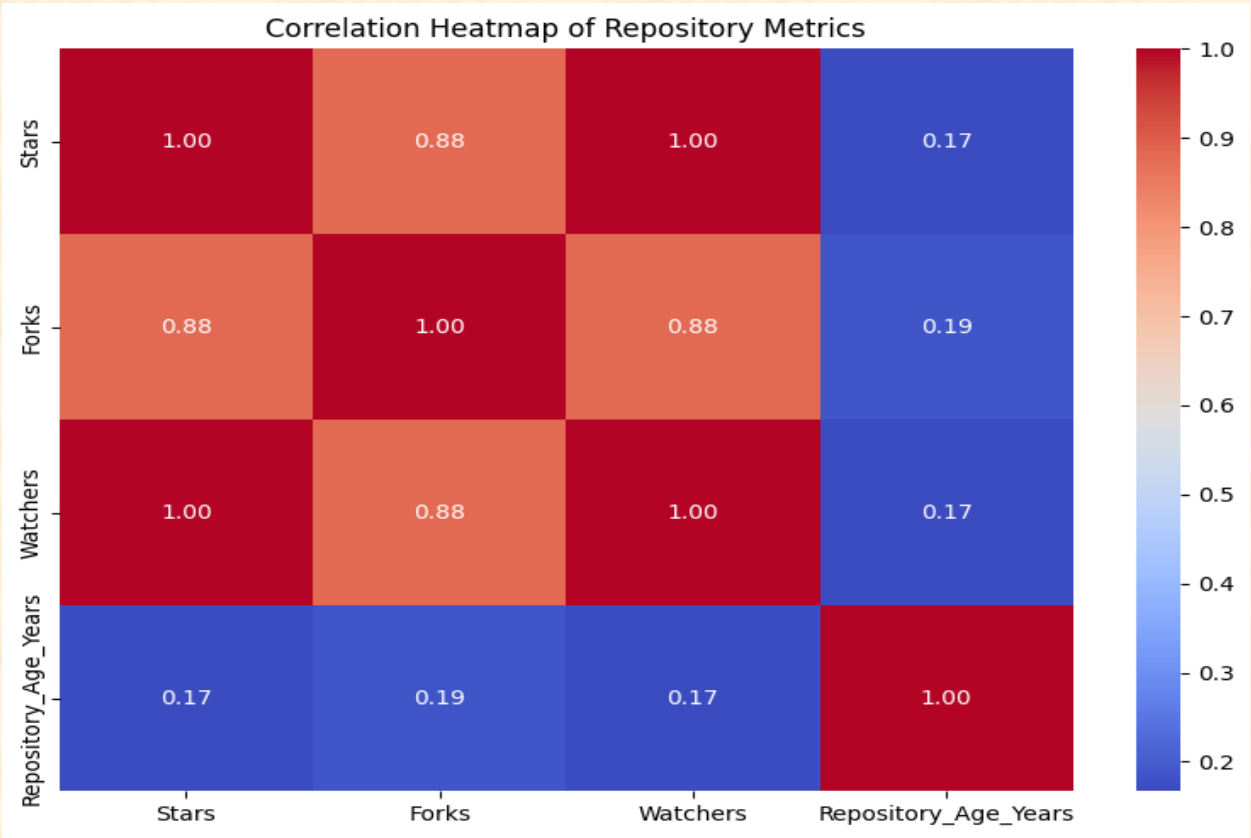
Correlation Heatmap

Columns Used

- Stars
- Forks
- Watchers
- Repository\_Age\_Years

Relationship

Repository Metrics ↔ Correlation



Interpretation

- 🌐 The heatmap visualizes the correlation among Stars, Forks, Watchers, and Repository\_Age\_Years.

- 🧠 Stars and Watchers exhibit a very strong positive correlation (1.00), indicating that repositories with more stars also tend to have more watchers.
- 🧠 Stars and Forks show a strong positive correlation (0.88), demonstrating that highly starred repositories are more likely to be forked by developers.
- 🧠 Repository Age shows only a weak positive relationship with Stars (0.17), Forks (0.19), and Watchers (0.17).
- 🧠 The correlation values range from -1 to +1, where values closer to +1 represent stronger positive relationships.
- 🧠 The analysis confirms that repository popularity and developer engagement metrics are strongly interconnected, whereas repository age has only a limited influence on popularity.

## 2. Average Stars and Forks by Company

### Chart Type

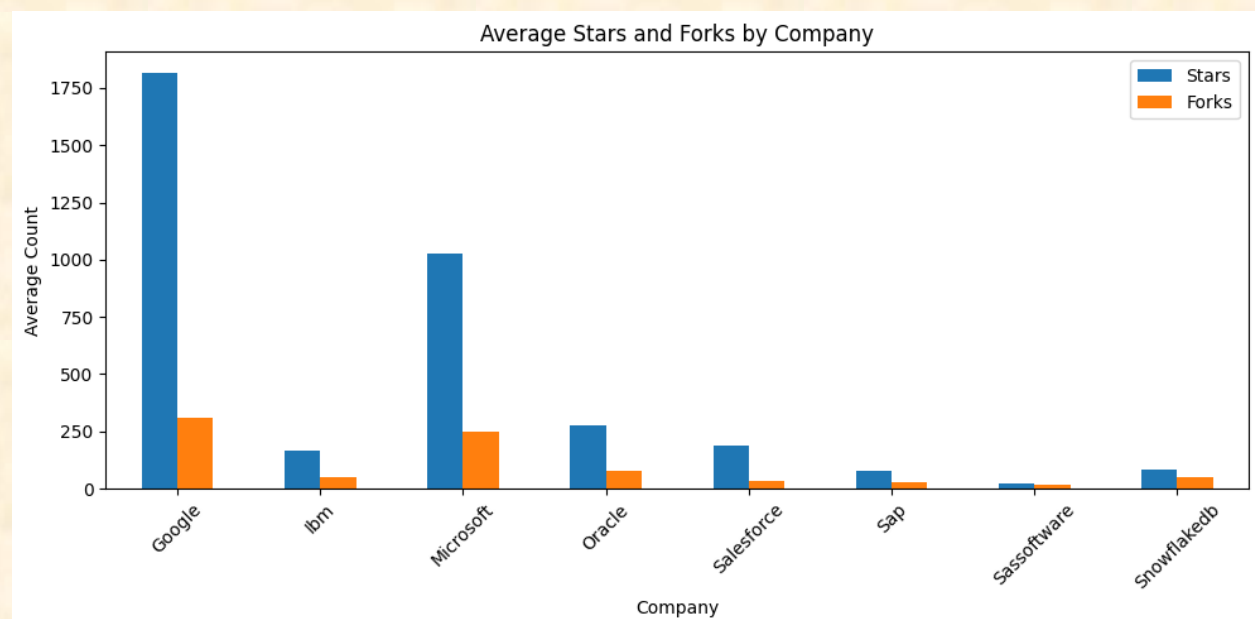
Grouped Bar Chart(From Pivot Table)

### Columns Used

- Company
- Stars
- Forks

### Relationship

Company ↔ Average Stars and Forks



## Interpretation

- The grouped bar chart compares the average Stars and average Forks across the selected technology companies.
- Google recorded the highest average Stars (1,816.86) and Forks (307.31), indicating the strongest developer engagement.
- Microsoft ranked second with 1,025.90 average Stars and 248.44 average Forks.
- Companies such as Oracle, IBM, Salesforce, SAP, Snowflake, and SAS recorded comparatively lower average Stars and Forks.
- The chart demonstrates that Google and Microsoft repositories consistently attract higher community attention, repository popularity, and developer contributions than the other companies.
- This visualization provides a clear comparison of open-source performance across organizations.
- Supports business decisions related to technology adoption and community engagement.

## 3. Pair Plot Analysis

### Chart Type

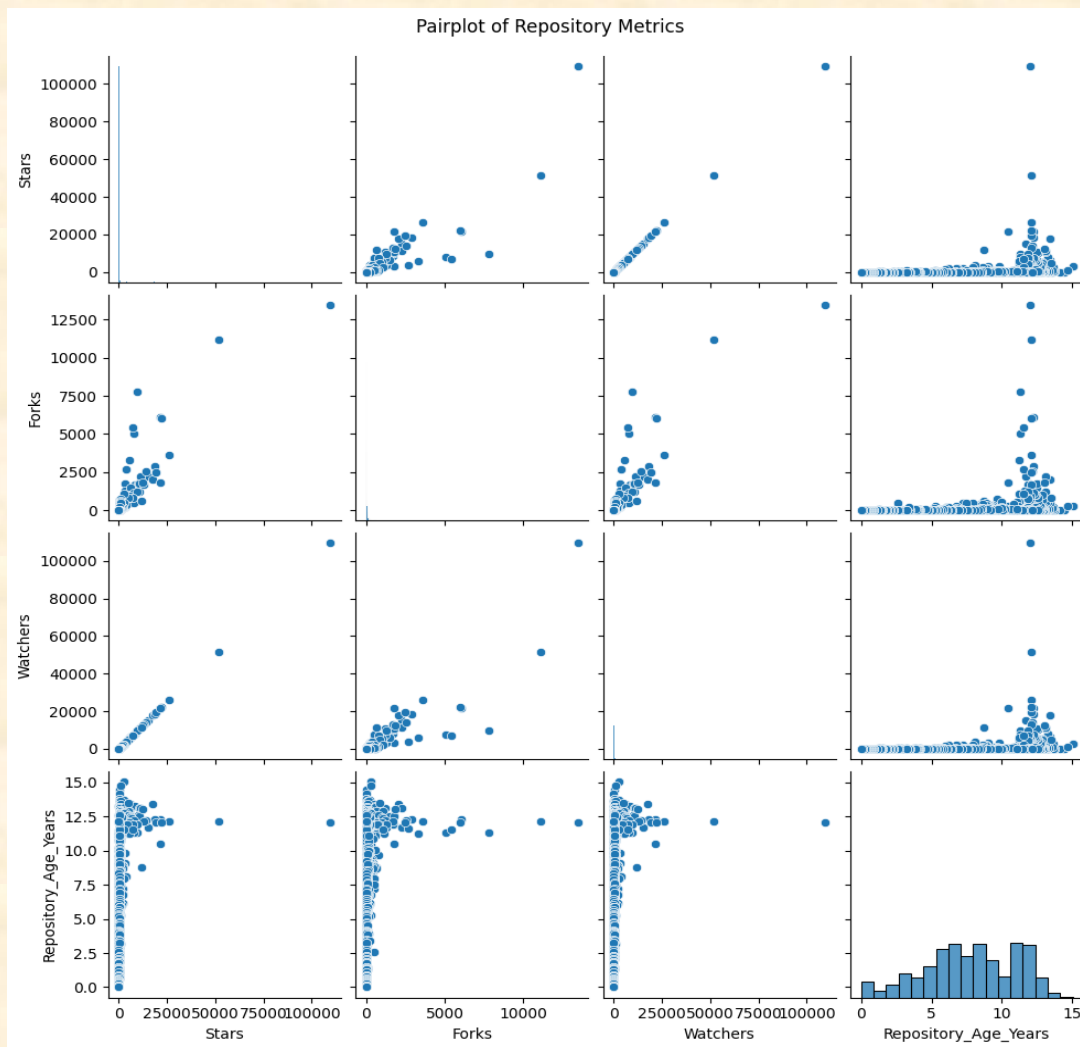
Pair Plot

### Columns Used

- Stars
- Forks
- Watchers
- Repository\_Age\_Years

### Relationship

Multiple Repository Metrics ↔ Relationships and Distributions

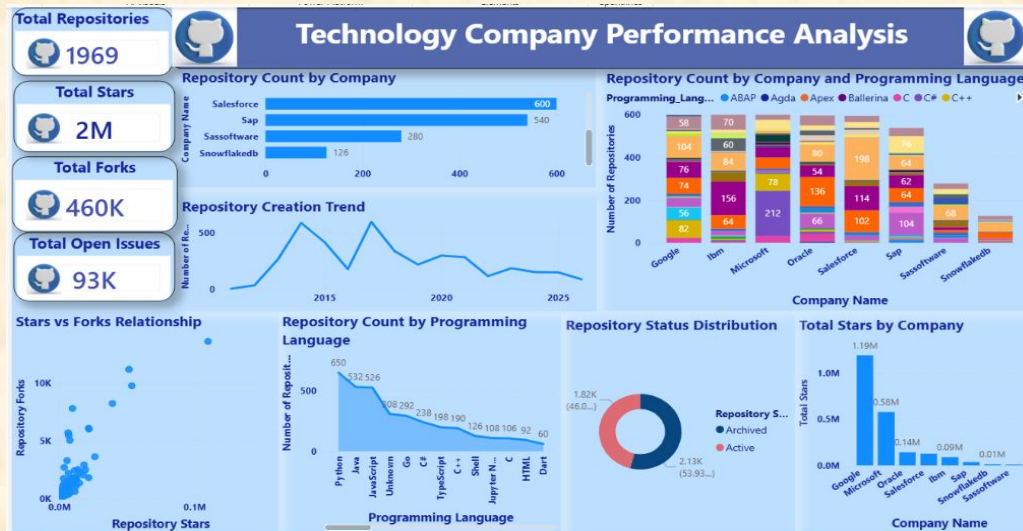


## Interpretation

- The pair plot analyzes relationships among Stars, Forks, Watchers, and Repository Age for 1,586 repositories.
- The average values are 520.83 Stars, 111.70 Forks, 520.83 Watchers, and 7.78 years of Repository Age.
- Repository stars range from 0 to 109,490, while forks range from 0 to 13,468.
- Repository Age varies from 0.00 to 15.05 years, indicating a combination of newly created and mature repositories.
- The pair plot shows that repositories with higher Stars generally receive higher Forks, while Stars and Watchers exhibit almost identical patterns.
- This multivariate analysis demonstrates that repository popularity and developer engagement vary significantly across repositories and helps identify relationships among multiple repository metrics.



## Dashboard:



## Key Insights:

- The GitHub repository dataset containing 1,586 repositories from eight leading technology companies was successfully analyzed using Python.
- Data preprocessing, exploratory data analysis (EDA), statistical analysis and data visualization were performed to evaluate repository popularity, developer engagement and technology trends.
- Google recorded the highest average repository popularity with 1,816.86 average Stars, followed by Microsoft with 1,025.90 average Stars, while Microsoft had the highest individual repository with 109,490 Stars.
- JavaScript, Python and Java were identified as the most widely used programming languages, highlighting their dominance in open-source software development.
- Repository Age ranged from 0.00 to 15.05 years, with most repositories between 5 and 12 years, indicating that the majority are mature and well-established projects.
- Repository Status analysis showed that 56.6% of repositories were Archived, while 43.4% remained Active, providing insights into repository maintenance patterns.
- Correlation analysis revealed a very strong positive relationship between Stars and Watchers (1.00) and a strong positive relationship between Stars and Forks (0.88), while Repository Age showed only a weak positive correlation (0.17) with repository popularity.
- Overall, the project successfully identified repository performance patterns, developer engagement, programming language trends, technology adoption, providing valuable insights to support data-driven decision-making.

## Types of Analysis:

### Descriptive Analysis

- Google achieved the highest average repository stars (1,816.86).

- 🌐 Microsoft recorded the highest maximum stars (109,490).
- 🌐 JavaScript, Python and Java were the most commonly used programming languages.
- 🌐 Most repositories were between 5–12 years old.

### **Diagnostic Analysis**

- 🌐 Strong positive correlation exists between Stars and Watchers (1.00).
- 🌐 Repository popularity is strongly associated with Forks (0.88).
- 🌐 Repository Age contributes only slightly to repository popularity (0.17 correlation).
- 🌐 Google and Microsoft repositories consistently demonstrate higher developer engagement than repositories from other companies.

### **Predictive Analysis**

- 🌐 Predict repository popularity based on repository characteristics.
- 🌐 Forecast future repository stars and forks.
- 🌐 Predict repository maintenance status.
- 🌐 Estimate developer engagement using repository metrics.
- 🌐 Forecast technology adoption trends across programming languages.

### **Prescriptive Analysis**

- 🌐 Encourage development using widely adopted programming languages where appropriate.
- 🌐 Improve maintenance of active repositories through regular updates and issue resolution.
- 🌐 Increase developer engagement by enhancing documentation and community participation.
- 🌐 Monitor repositories with declining activity to determine whether maintenance or archival is appropriate.
- 🌐 Use repository performance metrics to guide technology investments and open-source strategy.
- 🌐 Focus on successful repository characteristics to improve future project outcomes.

These recommendations can help technology companies strengthen their open-source ecosystem and improve developer collaboration.

## Future Enhancements

- 🔗 Expand the dataset by collecting repository data from additional technology companies.
- 🔗 Integrate real-time GitHub API data collection for continuous analysis.
- 🔗 Develop machine learning models to predict repository popularity and developer engagement.
- 🔗 Build interactive dashboards using Power BI, Tableau, or Looker Studio.
- 🔗 Automate data collection and reporting using workflow automation tools such as n8n.
- 🔗 Perform sentiment analysis on repository issues and discussions.
- 🔗 Include additional GitHub metrics such as Pull Requests, Contributors, Releases, and Commit History.
- 🔗 Develop a web-based analytics platform to provide real-time repository insights.

## Conclusion

- 🔗 This project successfully analyzed the open-source repository performance of leading technology companies using Python and the GitHub REST API.
- 🔗 Repository data from 1,586 repositories belonging to eight technology companies was collected, preprocessed, analyzed and visualized to generate meaningful business insights.
- 🔗 The analysis demonstrated the effectiveness of Python libraries such as Pandas, NumPy, Matplotlib and Seaborn for performing data preprocessing, statistical analysis, exploratory data analysis and visualization.
- 🔗 Repository metrics including Stars, Forks, Watchers, Repository Age, Programming Language, and Repository Status were examined to evaluate repository popularity, maintenance activity and developer engagement.
- 🔗 The results revealed that Google and Microsoft consistently achieved higher repository popularity and community engagement than the other selected companies.
- 🔗 Programming language analysis showed that JavaScript, Python, and Java remain the most widely adopted technologies in open-source software development.
- 🔗 Correlation analysis confirmed strong relationships between repository popularity and developer engagement metrics, while repository age had only a limited influence on popularity.
- 🔗 Overall, this project demonstrates how data analytics techniques can transform publicly available GitHub repository data into meaningful business insights.
- 🔗 The findings can support technology companies, researchers, and software development teams in understanding open-source trends, improving repository management, increasing developer engagement and making informed strategic decisions based on data.